

20 In solution, ammonium ions and cyanate ions react slowly to produce urea.



Keeping the temperature constant, different concentrations of $\text{CNO}^-(\text{aq})$ and $\text{NH}_4^+(\text{aq})$ are mixed and the relative initial rates of reaction determined.

experiment	$[\text{CNO}^-]$ / mol dm^{-3}	$[\text{NH}_4^+]$ / mol dm^{-3}	relative rate
1	0.5	2.5	5.80
2	0.5	0.5	1.16
3	2.0	2.5	23.20
4	1.0	2.0	X

What is X?

A 4.64

B 6.96

C 9.28

D 11.60